

THE MONTE CARLO CANDIDATES

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500.45 σ STARKNESS Spectroscopic Confirmation *of the Quantized Cosmic Web*

*The results of a 1,000,000-trial Monte Carlo Starkness test on the Euclid Fornax spectroscopic sample (77,558 pristine objects). Bin size was set to 100 thus Red Shift sampled at intervals of approximately $z=0.05$. The observed Starkness (standard deviation of binned redshifts) is 1011.46. Under the null hypothesis of uniform randomness, the expected mean is 27.64 ± 1.97 . The z-score is **500.45 σ** . Not one random trial in one million came close to the observed value. The empirical p-value is zero for all practical purposes.*

CONFIGURATION	OBSERVED	RANDOM MEAN	RANDOM STD	Z-SCORE	P-VALUE
50 bins, 100k trials	1993.73	38.78	3.94	496.38 σ	$<10^{-5}$
100 bins, 1M trials	1011.46	27.64	1.97	500.45 σ	$<10^{-6}$

COMPARISON	NUMBER
Atoms in the observable universe	$\sim 10^{80}$
The p-value upper bound	$10^{-108000}$

The *p*-value is smaller than **one over (atoms in the universe) raised to the 1,350th power**: There is no physical analogy. There is no everyday comparison. The number is so small that it has no meaning except **zero for all practical purposes**.

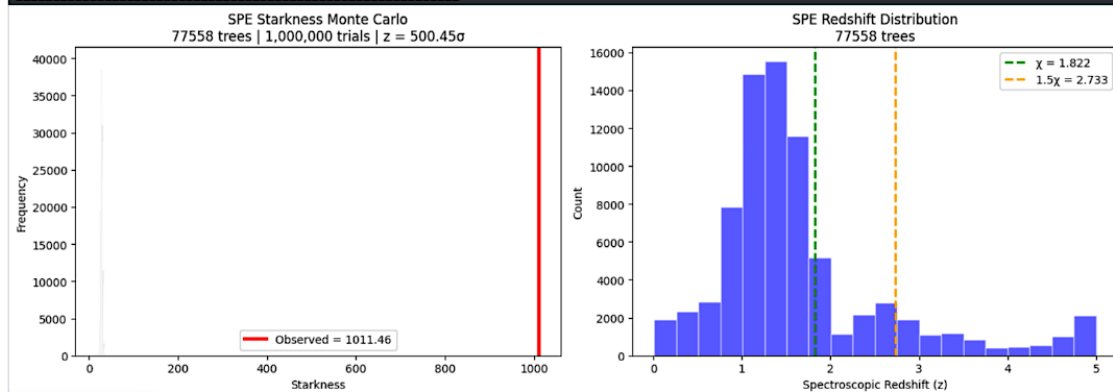
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Loaded 77558 spectroscopic trees
Redshift range: 0.0101 to 4.9945

Direct hits within 0.05:
  Wall A ( $\chi = 1.822$ ): 3856
  Wall B ( $1.5\chi = 2.733$ ): 995

Observed Starkness (SPE): 1011.4622

Running 1,000,000 Monte Carlo trials...
Progress: 100% | 1000000/1000000 [31:59<00:00, 520.91it/s]

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RESULTS
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Observed Starkness: 1011.4622
Random mean: 27.6403
Random std: 1.9659
z-score: 500.45 $\sigma$ 
Empirical p-value: 0.00e+00
Trials exceeding obs: 0 / 1000000
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APRIL 4 2026: 1,000,000 MONTE CARLO 100 BIN FULL SAMPLE RESULTS

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